# Business Question and Visualization Report

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| Date | 28 October 2026 |
| Team ID | EN25CS3010655 |
| Project Name | Health360: Comprehensive Healthcare Risk & Disease Analytics |
| Maximum Marks | 5 Marks |

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

## Business Questions and Visualisation

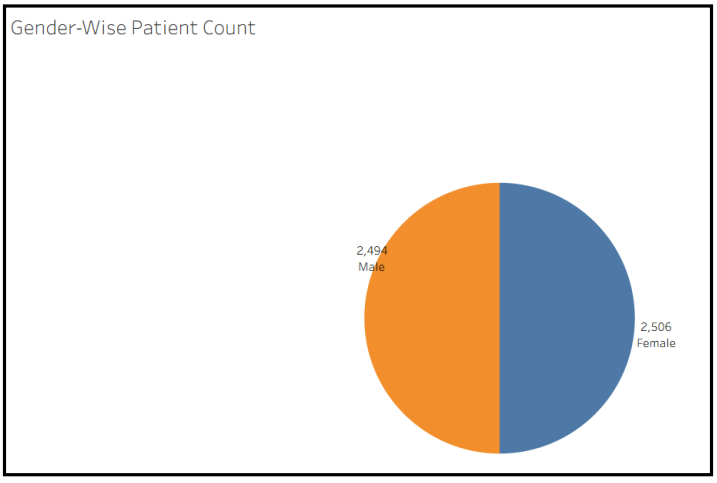
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

Business Questions and Visualisations (Health360 Analytics)

**1. What is the overall distribution of patient disease risk levels across the monitored population?**

```
SELECT
   risk_level,
   COUNT(patient_id) AS total_patients,
   ROUND(COUNT(patient_id) * 100.0 / SUM(COUNT(patient_id)) OVER(), 2) AS percentage
FROM HEALTH360_ANALYTICS.PATIENT_RISK_ASSESSMENT
GROUP BY risk_level
ORDER BY total_patients DESC;
```

**Screenshot / Description of Visualisation:**



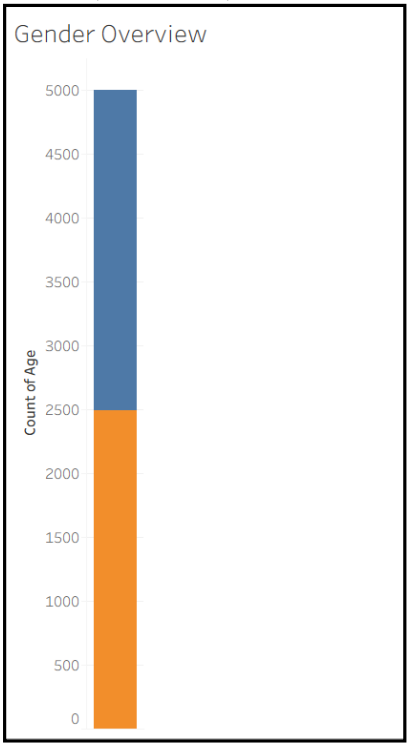
**Visualization Overview:** The pie chart *Gender-Wise Patient Count* splits the monitored patient population by gender. It describes the composition of the population on which the disease risk levels are assessed.

**Key Observations:** The chart shows 2,506 female patients and 2,494 male patients, which is 5,000 patients in total. Females make up about 50.1% and males about 49.9%, a difference of only 12 patients, so the two halves of the pie are almost identical.

**Business Insights:** The monitored population is evenly balanced between genders. Risk levels and disease patterns calculated for the Health360 population are therefore not dominated by one gender, and risk can be compared between male and female patients without any adjustment for group size.

**Conclusion:** The patient base is split almost equally by gender, which gives a fair and reliable baseline for the risk distribution analysis.

**Screenshot / Description of Visualisation (continued):**




**Visualization Overview:** The stacked bar chart *Gender Overview* shows the total patient count (Count of Age) as one bar divided into two colour segments. The colours match the pie chart above: blue for female and orange for male.

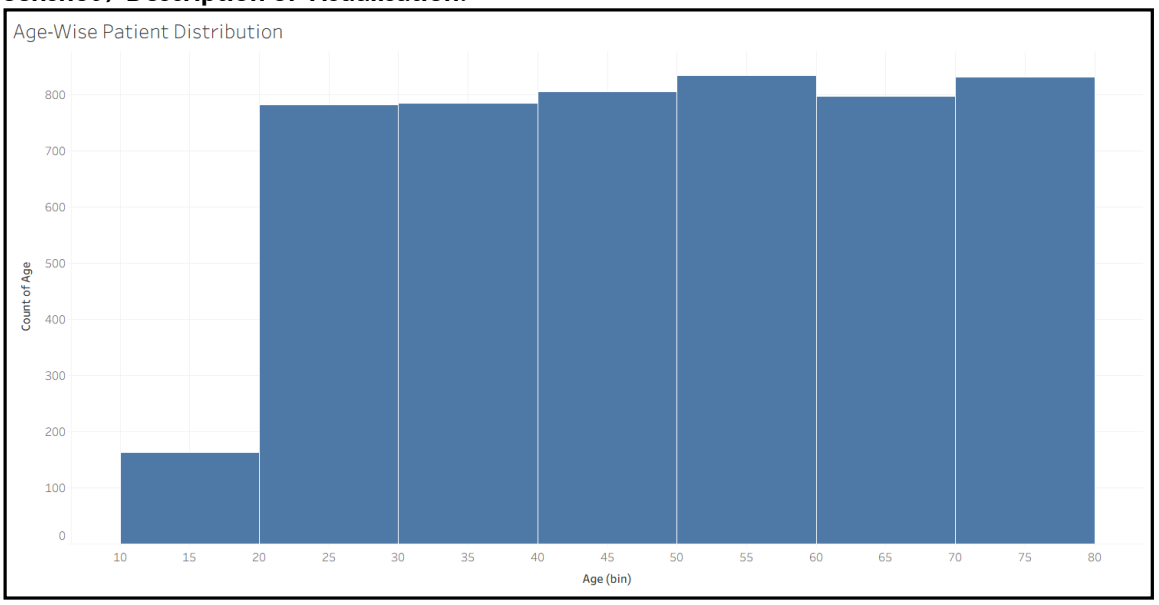
**Key Observations:** The bar reaches 5,000 on the vertical axis. The orange segment runs from 0 to about 2,500 and the blue segment from about 2,500 to 5,000, so each gender forms about half of the bar.

**Business Insights:** The chart confirms that the dataset contains 5,000 patient records and that both genders are equally represented. This gives the dashboard a clear overall population count that can be filtered by gender when risk levels are explored.

**Conclusion:** The total population is 5,000 patients and it is divided evenly between genders, which supports the gender-wise comparison of health risk.

**2. What are the monthly heart rate and SpO2 anomaly trends over time?**

**Screenshot / Description of Visualisation:**




**Visualization Overview:** The histogram *Age-Wise Patient Distribution* shows how many patients fall into each 10-year age bin from 10 to 80 years. Although the question concerns monthly vital-sign trends, this view gives the age profile of the patients being monitored.

**Key Observations:** The 10–20 bin is the smallest, with about 160 patients. Every bin from 20 to 80 holds roughly 780 to 835 patients. The highest bars are the 50–60 and 70–80 bins (about 830 patients each), while the 20–30, 30–40, 40–50 and 60–70 bins are close behind.

**Business Insights:** Most monitored patients are adults, and about half of them are aged 50 or above. Older patients generally need closer monitoring for chronic conditions, so heart rate and SpO2 alerts in the Health360 system will mostly relate to middle-aged and elderly patients. A very small share of patients is under 20.

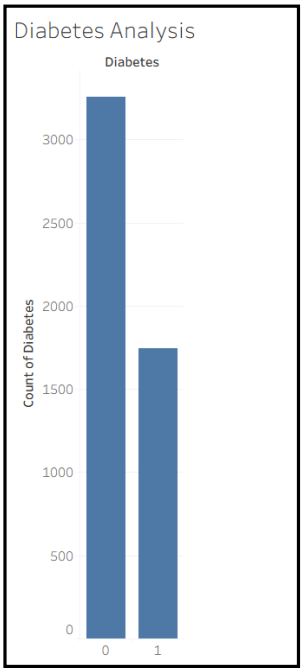
**Conclusion:** The patient population is spread fairly evenly across adult age groups, with a large older segment that should receive priority in monitoring and early-warning efforts.

**3. Which age groups exhibit the highest prevalence of cardiovascular and diabetic risk factors?**

**Code**

```
SELECT
  CASE
    WHEN age < 30 THEN '18-29'
    WHEN age BETWEEN 30 AND 45 THEN '30-45'
    WHEN age BETWEEN 46 AND 60 THEN '46-60'
    ELSE '60+'
  END AS age_group,
  COUNT(CASE WHEN primary_diagnosis = 'Cardiovascular' THEN 1 END) AS cardio_cases,
  COUNT(CASE WHEN primary_diagnosis = 'Diabetes' THEN 1 END) AS diabetes_cases
FROM HEALTH360_ANALYTICS.PATIENT_DEMOGRAPHICS_EHR
GROUP BY age_group
ORDER BY age_group;
```

**Screenshot / Description of Visualisation:**



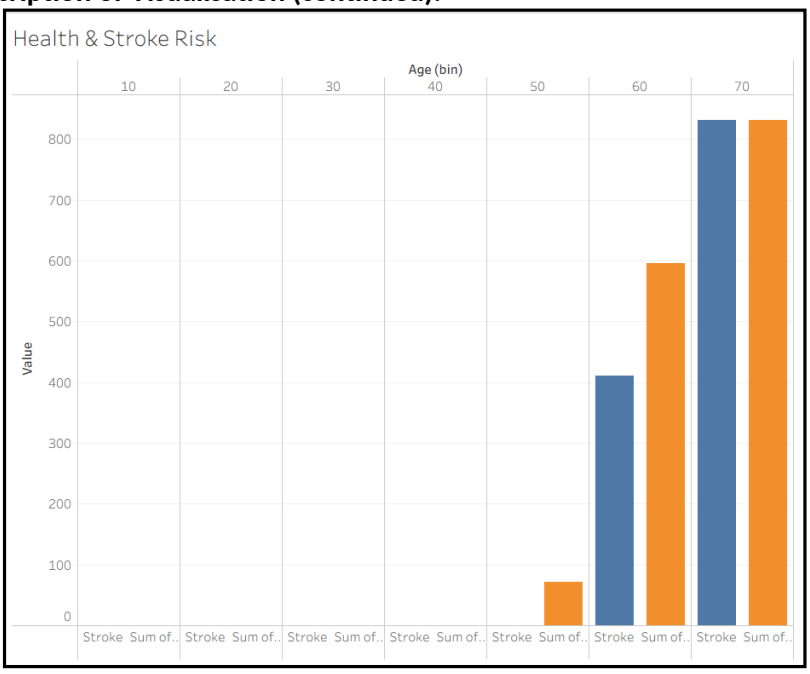

**Visualization Overview:** The bar chart *Diabetes Analysis* shows the number of patients by diabetes status, where 0 means the patient does not have diabetes and 1 means the patient has diabetes.

**Key Observations:** About 3,250 patients are in the non-diabetic group (0) and about 1,750 are in the diabetic group (1), out of 5,000 patients. This means roughly 65% of patients are non-diabetic and 35% are diabetic. The non-diabetic bar is almost twice as tall as the diabetic bar.

**Business Insights:** About one in every three monitored patients has diabetes, which is a large share of the population and a major risk factor for cardiovascular complications. Health360 should therefore treat diabetes screening, glucose monitoring and follow-up as a priority.

**Conclusion:** Diabetes affects roughly 35% of the patients, so it is one of the most important chronic conditions to track in the analytics project.

**Screenshot / Description of Visualisation (continued):**




**Visualization Overview:** The grouped bar chart *Health & Stroke Risk* compares two measures across 10-year age bins (10 to 70): Stroke (blue) and a second measure labelled "Sum of..." (orange), whose full name is cut off in the screenshot.

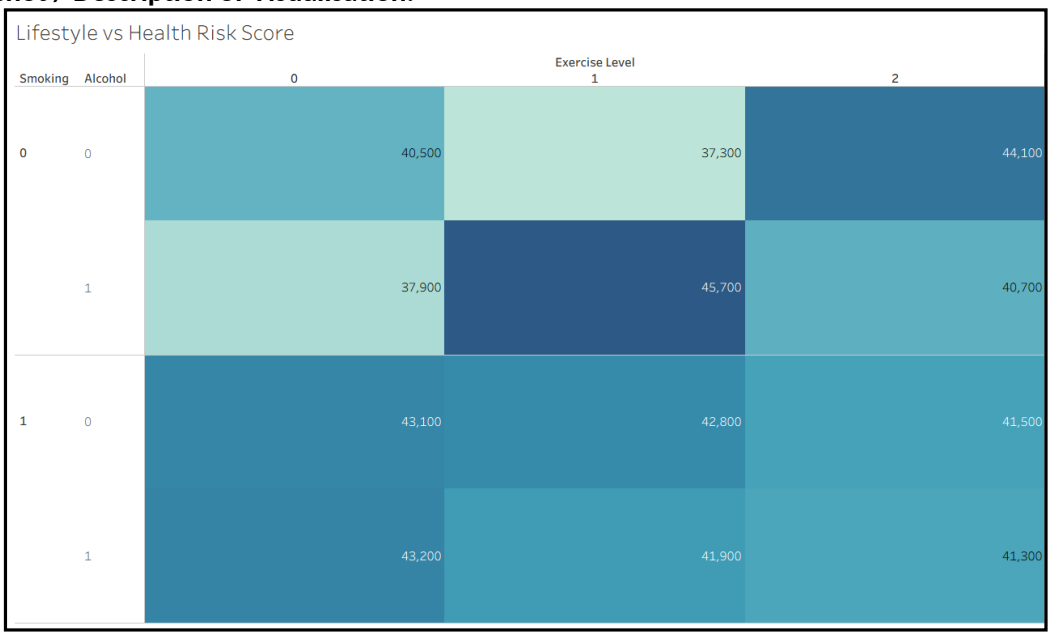
**Key Observations:** No bars are visible for the 10, 20, 30 and 40 age bins. In the 50 bin only the orange measure appears (about 70). In the 60 bin the stroke bar is about 410 and the orange bar about 595. In the 70 bin both bars reach about 830. Both measures therefore increase sharply with age.

**Business Insights:** Risk is concentrated in older patients. Stroke cases appear only from the 60 age bin onwards, and the other measure begins in the 50 bin. Patients aged 50 and above should be the main target group for preventive screening and regular monitoring of cardiovascular and related risk factors.

**Conclusion:** Health risk, including stroke, rises steeply after age 50, so age is a key factor for prioritising patients in the Health360 risk assessment.

**4. How do emergency risk alerts vary by hospital department / ward?**

**Screenshot / Description of Visualisation:**




**Visualization Overview:** The heat map *Lifestyle vs Health Risk Score* shows the aggregated health risk score for every combination of smoking (0/1), alcohol (0/1) and exercise level (0, 1, 2). Darker blue cells represent higher values. It supports the risk-alert question by showing which lifestyle profiles carry higher risk scores.

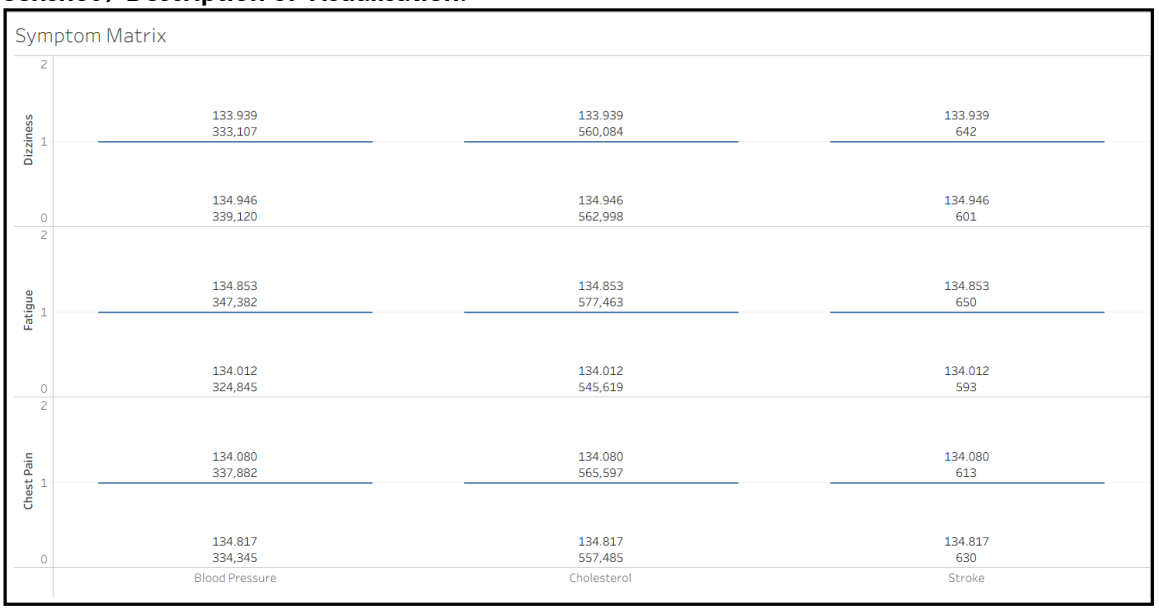
**Key Observations:** The highest value is 45,700 (non-smoker, alcohol use, exercise level 1), followed by 44,100 (non-smoker, no alcohol, exercise level 2). The lowest values are 37,300 (non-smoker, no alcohol, exercise level 1) and 37,900 (non-smoker, alcohol use, exercise level 0). All smoker combinations lie in a narrow band of 41,300 to 43,200, and for smokers the value falls as the exercise level rises from 0 to 2, both with and without alcohol.

**Business Insights:** Smoking patients consistently show moderate-to-high values, and higher exercise levels are linked with lower values among smokers. Among non-smokers the pattern is mixed. Since the cells are aggregate values, they may also reflect group sizes, so lifestyle-based alerts should be supported by further analysis.

**Conclusion:** Lifestyle habits influence health risk, and encouraging exercise, especially for smokers, is a practical way to reduce risk and the alerts it generates.

**5. What is the average patient readmission rate grouped by primary diagnosis?**

**Screenshot / Description of Visualisation:**




**Visualization Overview:** The symptom matrix *Symptom Matrix* compares three reported symptoms (Dizziness, Fatigue and Chest Pain), each shown as absent (0) or present (1), against Blood Pressure, Cholesterol and Stroke. Each cell shows an aggregate value and a second value of about 134. It relates to diagnosis-level outcomes by showing how a serious condition (stroke) differs across symptom groups.

**Key Observations:** Stroke counts are higher for patients with dizziness (642 vs 601 without) and with fatigue (650 vs 593 without), but slightly lower for patients with chest pain (613 vs 630 without). Blood pressure and cholesterol totals are of similar size in the symptom and no-symptom groups. The second value stays between about 133.9 and 134.9 in every row.

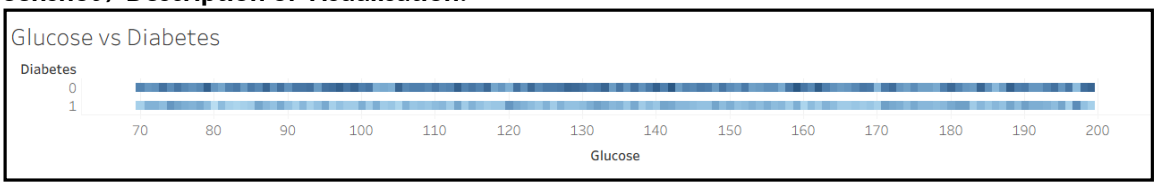
**Business Insights:** Dizziness and fatigue are associated with slightly more stroke cases, while chest pain does not show the same increase. The differences are small, so symptoms alone are weak indicators of serious outcomes. Because the totals are sums, group sizes also affect them. Symptoms should be combined with age, blood pressure and other health measures.

**Conclusion:** Reported symptoms give only limited separation between patient groups and should be used together with clinical measurements for reliable risk identification.

**6. How does real-time sensor device latency perform across different IoT gateway connection protocols?**

```
SELECT
   connection_protocol,
   AVG(data_latency_ms) AS avg_latency_ms,
   MAX(data_latency_ms) AS max_latency_ms,
   COUNT(sensor_id) AS total_readings
FROM HEALTH360_ANALYTICS.IOT_SENSOR_TELEMETRY
GROUP BY connection_protocol
ORDER BY avg_latency_ms ASC;
```

**Screenshot / Description of Visualisation:**




**Visualization Overview:** The heat map *Glucose vs Diabetes* shows glucose values from about 70 to 200 along the horizontal axis for non-diabetic patients (row 0) and diabetic patients (row 1). Darker shading means more patients at that glucose value. Glucose is the type of measurement that monitoring devices stream in real time, so this view shows what those readings reveal about diabetes status.

**Key Observations:** Both rows cover the whole range from about 70 to 200. The non-diabetic row (0) is darker overall, which fits with its larger group of about 3,250 patients, while the diabetic row (1) is lighter but spread fairly evenly over the same range. No glucose range belongs only to diabetic patients.

**Business Insights:** Glucose alone does not clearly separate diabetic from non-diabetic patients in this data, because both groups have readings across the full range. Health360 should therefore combine glucose with other factors such as age, BMI and blood pressure, and should deliver glucose readings promptly so that high values can be reviewed quickly.

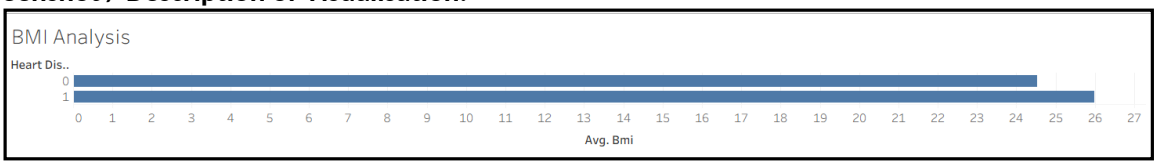
**Conclusion:** Glucose levels overlap heavily between the two groups, so a single glucose threshold is not enough to identify diabetes risk.

**7. What is the correlation between body mass index (BMI), blood pressure, and high-risk health outcomes?**

**Code (Snowflake SQL Query):**

```
SELECT
   ROUND(bmi, 0) AS bmi_rounded,
   AVG(systolic_bp) AS avg_systolic,
   AVG(diastolic_bp) AS avg_diastolic,
   COUNT(CASE WHEN risk_level = 'High' THEN 1 END) AS high_risk_count
FROM HEALTH360_ANALYTICS.PATIENT_HEALTH_METRICS
GROUP BY bmi_rounded
HAVING COUNT(patient_id) > 10
ORDER BY bmi_rounded ASC;
```

**Screenshot / Description of Visualisation:**




**Visualization Overview:** The bar chart *BMI Analysis* shows the average BMI for patients without heart disease (0) and with heart disease (1). It addresses the BMI part of the question; blood pressure is not shown in this view.

**Key Observations:** The average BMI is about 24.5 for patients without heart disease and about 26 for patients with heart disease. Patients with heart disease therefore have an average BMI that is about 1.5 points higher.

**Business Insights:** The heart disease group has an average BMI above 25, which is in the overweight range, while the other group is just below it. This suggests that higher body weight is related to heart disease risk. BMI is a simple routine measure that Health360 can use to flag patients who may benefit from weight management and preventive cardiac care.

**Conclusion:** Patients with heart disease have a higher average BMI, so BMI is a useful indicator for high-risk health outcomes.

**8. What are the top 5 predicted chronic conditions projected to increase over the next 6 months?**

**Code (Snowflake SQL Query):**

```
SELECT
   predicted_disease,
   SUM(predicted_cases) AS projected_cases,
   AVG(model_confidence_score) AS avg_confidence
FROM HEALTH360_ANALYTICS.PREDICTIVE_DISEASE_FORECAST
WHERE forecast_period = 'Next 6 Months'
GROUP BY predicted_disease
ORDER BY projected_cases DESC
LIMIT 5;
```